Detecting Clickbait and Sensationalism in Ghanaian Online News Headlines

A new labelled dataset and a comparison of classical ML, a fine-tuned transformer and zero-shot classification

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github.com/Mackey-Kumi/ghana-clickbait-detection

Ghanaian clickbait has its own style

Real headlines from the dataset

"My Manager Wanted To Rape Me – Ruby Of Date Rush"
quote with no facts about what happened

"Kofi Kinaata Puts His Huge House And Plush Living Room On Display (Watch Video)"
vague hype, nothing checkable

"Davido Spoils Bride Chioma with $300K Richard Mille Diamond Watch at Miami Wedding"
same topic, but informative: every claim is checkable

Research questions

1. How well does a simple TF-IDF + Logistic Regression model work?
2. Does fine-tuning DistilBERT do noticeably better?
3. Can a zero-shot model with no training do the job?
4. Do models learn clickbait style, or just which outlet a headline came from?

1,000 headlines from seven Ghanaian outlets

1,000

headlines, scraped 25 Aug 2026

500 / 500

mainstream vs. gossip outlets

19.6%

labelled Clickbait (196); 79.4% Informative, 1% Ambiguous

Mainstream

Myjoyonline 155 · Citinewsroom 155 · GNA 155 · Graphic Online 35
Only 15 of 500 headlines (3%) labelled Clickbait

Entertainment / gossip

GhPage 167 · ZionFelix 167 · Yen.com.gh 166
Only 181 of 500 (36%) labelled Clickbait, so a gossip site ≠ a clickbait headline

Ethics: robots.txt respected (2 sites excluded), 1.5 s between requests, headline text and URL only

Labelling: AI drafts, every label checked by hand

167

of 1,000 AI draft labels overridden (16.7%)

32 / 196

clickbait headlines the AI caught on its own

Three codebook revisions while labelling

1. **Disturbing ≠ clickbait**
   A grim story that says who did what is informative. The test is whether information is withheld.

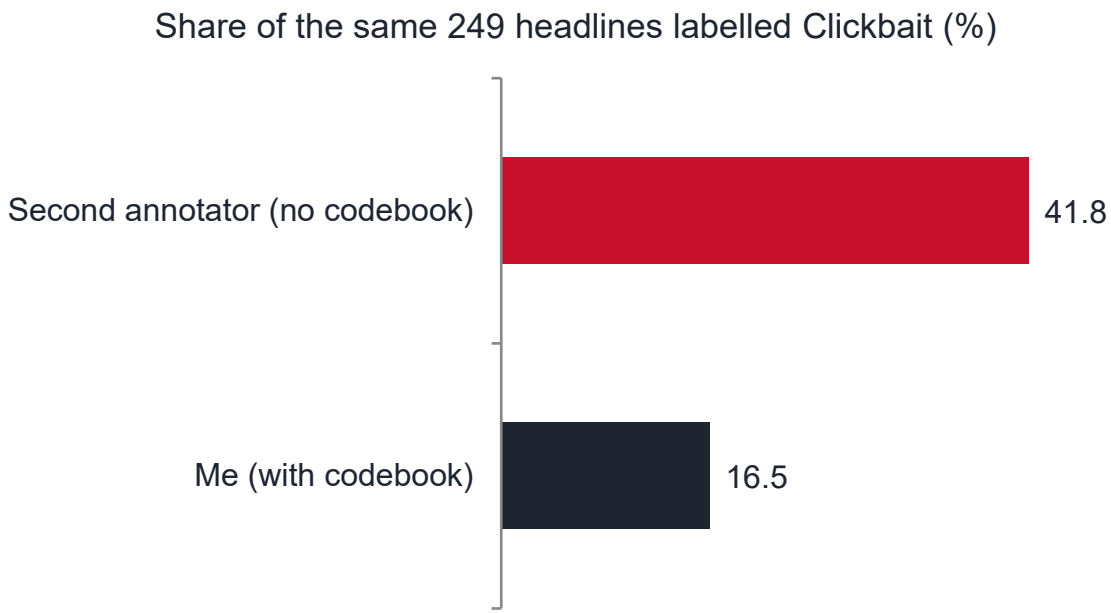
2. **A quote is not automatically informative**
   The framing must add a fact the quote alone doesn't give.

3. **Vague hype is not description**
   “Huge”, “plush”, “stunning” with nothing checkable = curiosity gap.

A second annotator mostly disagreed with me

κ = 0.057

Cohen's kappa on 250 blind headlines: “slight” agreement (raw agreement 54.6%)



Why: they labelled without seeing the codebook, so they used a broad everyday meaning (“any gossip story”). Of the 205 headlines I called Informative, they called 86 Clickbait. So “clickbait” isn't well defined until it is written down, and the labels depend on annotators being trained on it.

Three approaches, one shared test set

1

Baseline

TF-IDF (1–2-grams) + Logistic Regression, class-balanced

Fast and interpretable; runs on a laptop

2

Fine-tuned DistilBERT

distilbert-base-uncased, class-weighted loss, 4 epochs

Trained on a free Kaggle T4 GPU in ~14 s

3

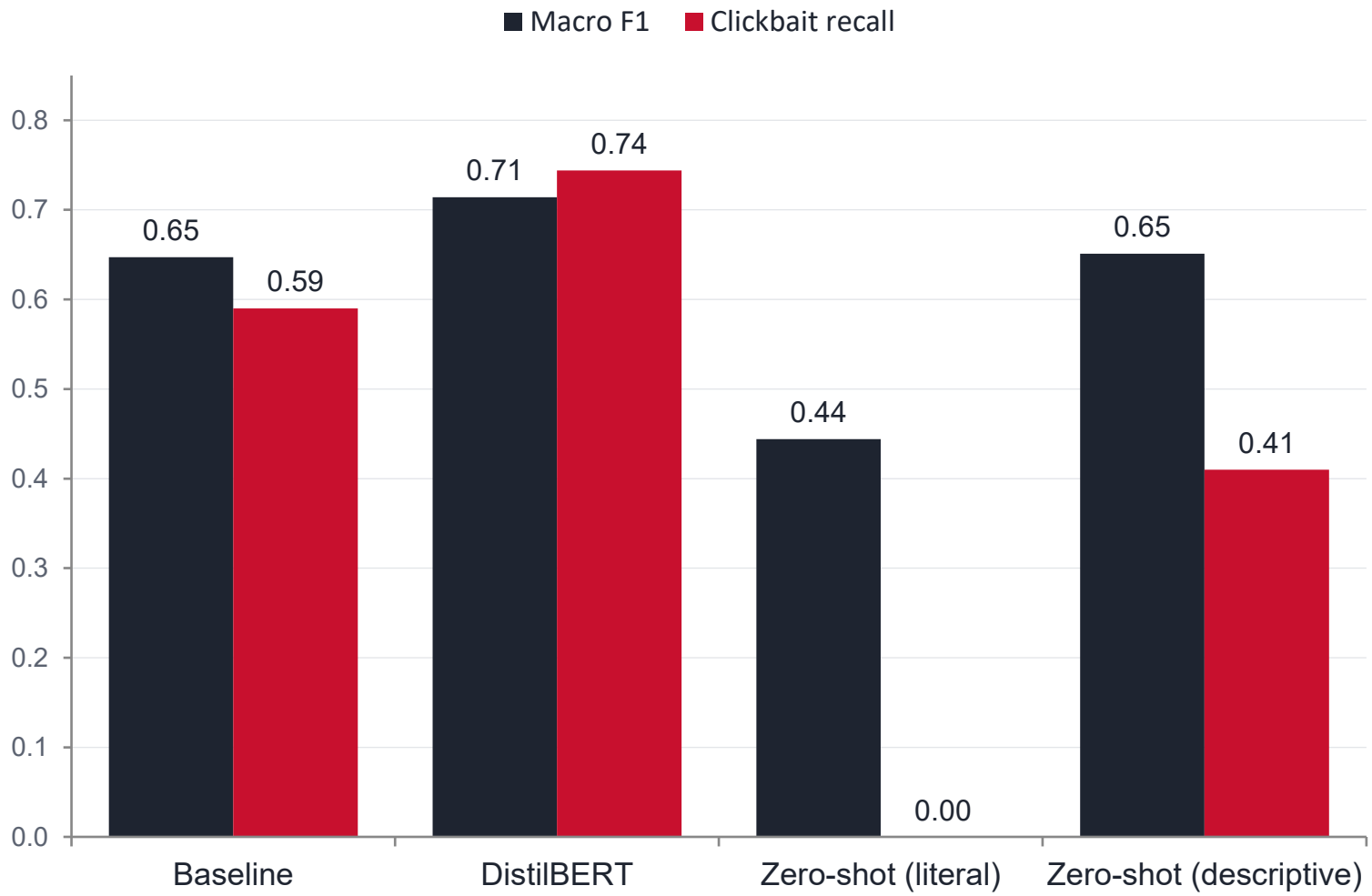
Zero-shot

facebook/bart-large-mnli, no training on my data

Two label wordings tested

- 990 headlines (Ambiguous dropped), stratified 80/20 split, seed 42: same 198 test headlines for every model
- Main metrics: macro F1 and Clickbait recall. Accuracy misleads at 80/20 (“always Informative” scores ~80%)
- Held-out-outlet test: train on 6 outlets, test on the unseen 7th (answers RQ4)

Fine-tuned DistilBERT wins where it matters



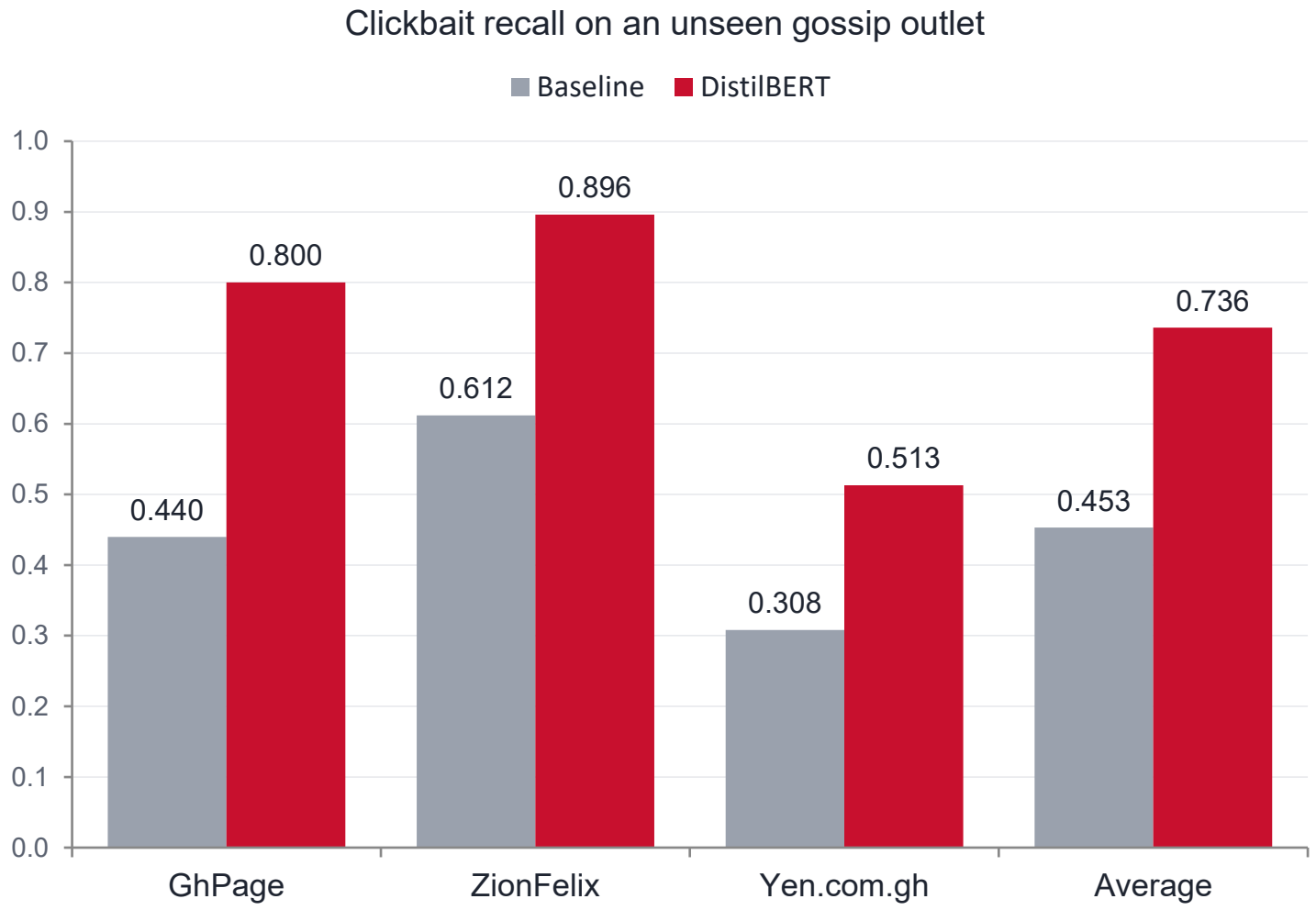
29 / 39

clickbait test headlines caught by DistilBERT (baseline 23, zero-shot 16)

0.798

highest accuracy in the project, from the model that caught no clickbait at all

Only DistilBERT holds up on outlets it never saw



Baseline

0.590 → 0.453

Leans on outlet and topic words: mahama, npp, celebrity names

DistilBERT

0.744 → 0.736

Barely drops: closer to learning clickbait style itself

Zero-shot results depend on how the labels are worded

Literal labels

“clickbait” / “informative”

0.000

Clickbait recall: never predicted clickbait, on any of the 7 outlets

Descriptive labels

“a sensationalized clickbait headline designed to provoke curiosity” / “a factual, informative news headline”

0.410

Clickbait recall on the identical test set; macro F1 0.651, level with the trained baseline

Same model, same headlines; only the wording changed. A zero-shot result is a result about the model plus the prompt.

Error analysis: the baseline learns gossip vocabulary

36 false alarms

Informative headlines flagged as clickbait. 33 of 36 from gossip sites

- Factual celebrity news with “photos” / “video”: “Jackie Appiah Celebrates Son … Shares Lovely Photos”
- Short, grim but complete crime headlines: “Wife stabs younger husband to death”
- “(Watch Video)” tags on otherwise complete headlines

16 missed clickbait

All from gossip sites. Exactly the codebook edge cases

- Vague hype: “… Flaunts Her Fine Figure …”, “… Opens a Plush Restaurant …”
- Quotes with no anchoring fact: “Sleeping with another person … is not cheating – Okyeame Kwame”
- “BREAKING- Parliament passes …”: invisible, because TF-IDF lowercases everything

Limitations and responsible use

Limitations

- Small: 196 clickbait headlines, only 39 in the test set; no confidence intervals
- One primary annotator; the agreement check used an untrained second annotator
- Topic confound: clickbait is mostly entertainment, politics mostly informative
- One-day snapshot, headlines only, English only
- No per-headline predictions saved for DistilBERT and zero-shot

Ethics and AI use

- robots.txt respected: two sites excluded, not worked around
- No personal data; second annotator anonymised
- “Clickbait” judges style, not truth, and is not an outlet ranking
- Precision 0.47: fine for a warning label, never for blocking
- AI drafted labels and helped with code and writing; every label hand-checked, code tested, numbers checked against saved outputs

Conclusion

- **Dataset:** 1,000 hand-checked Ghanaian headlines and a Ghana-specific codebook
- **Best model:** Fine-tuned DistilBERT: macro F1 0.714, Clickbait recall 0.744, and it holds up on unseen outlets
- **Zero-shot:** Usable only with carefully worded labels; fails completely with the obvious ones
- **Hardest part:** Defining clickbait: three codebook revisions, $\kappa = 0.057$ without the codebook

Next: train a second annotator on the codebook · more data, especially clickbait from mainstream outlets · keep capitalisation as a feature · multilingual models for Twi/Pidgin

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